EIGRP

R1

```
0, changed state to up
*Oct 10 15:27:59.815: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
1, changed state to up
*Oct 10 15:27:59.815: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
2, changed state to up
*Oct 10 15:27:59.815: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
3, changed state to up
R1#
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#
*Oct 10 15:28:10.899: %LINK-3-UPDOWN: Interface Ethernet0/2, changed state to do
wn
*Oct 10 15:28:10.900: %LINK-3-UPDOWN: Interface Ethernet0/3, changed state to do
wn
*Oct 10 15:28:11.900: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
2, changed state to down
*Oct 10 15:28:11.901: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
3, changed state to down
R1(config)#router eigrp 10
R1(config-router)#network 10.0.0.0
R1(config-router)#network 20.0.0.0
R1(config-router)#exit
R1(config)#
*Oct 10 15:29:14.660: %DUAL-5-NBRCHANGE: EIGRP-IPv4 10: Neighbor 20.0.0.2 (Ether
net0/1) is up: new adjacency
R1(config)#
*Oct 10 15:33:00.564: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Confi
g Wizard)
R1(config)#do sh ip eigrp neighbor
EIGRP-IPv4 Neighbors for AS(10)
H   Address                 Interface              Hold Uptime   SRTT   RTO  Q  Seq
                                                   (sec)         (ms)       Cnt Num
0   20.0.0.2                Et0/1                    13 00:12:45    6   100  0  8
R1(config)#do sh ip eigrp topology
EIGRP-IPv4 Topology Table for AS(10)/ID(20.0.0.1)
Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - reply Status, s - sia Status

P 10.0.0.0/8, 1 successors, FD is 281600
        via Connected, Ethernet0/0
P 30.0.0.0/8, 1 successors, FD is 307200
        via 20.0.0.2 (307200/281600), Ethernet0/1
P 40.0.0.0/8, 1 successors, FD is 332800
        via 20.0.0.2 (332800/307200), Ethernet0/1
P 20.0.0.0/8, 1 successors, FD is 281600
        via Connected, Ethernet0/1
```

R2

```
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router eigrp 10
R2(config-router)#network 20.0.0.0
R2(config-router)#network 20.0.0.0
*Oct 10 15:29:14.665: %DUAL-5-NBRCHANGE: EIGRP-IPv4 10: Neighbor 20.0.0.1 (Ether
net0/0) is up: new adjacency
R2(config-router)#network 30.0.0.0
R2(config-router)#exit
R2(config)#
*Oct 10 15:29:52.895: %DUAL-5-NBRCHANGE: EIGRP-IPv4 10: Neighbor 30.0.0.2 (Ether
net0/1) is up: new adjacency
R2(config)#do sh ip eigrp neighbor
*Oct 10 15:33:13.884: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Confi
g Wizard)
R2(config)#do sh ip eigrp neighbor
EIGRP-IPv4 Neighbors for AS(10)
H   Address                 Interface              Hold Uptime   SRTT   RTO  Q
Seq
                                                   (sec)         (ms)       Cnt
Num
1   30.0.0.2                Et0/1                    14 00:03:22    5   100  0
4
0   20.0.0.1                Et0/0                    13 00:04:01    5   100  0
4
R2(config)#do sh ip eigrp topology
EIGRP-IPv4 Topology Table for AS(10)/ID(30.0.0.1)
Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - reply Status, s - sia Status

P 10.0.0.0/8, 1 successors, FD is 307200
        via 20.0.0.1 (307200/281600), Ethernet0/0
P 30.0.0.0/8, 1 successors, FD is 281600
        via Connected, Ethernet0/1
P 40.0.0.0/8, 1 successors, FD is 307200
        via 30.0.0.2 (307200/281600), Ethernet0/1
P 20.0.0.0/8, 1 successors, FD is 281600
        via Connected, Ethernet0/0
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```
R2(config)#do sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

D     10.0.0.0/8 [90/307200] via 20.0.0.1, 00:06:26, Ethernet0/0
      20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        20.0.0.0/8 is directly connected, Ethernet0/0
L        20.0.0.2/32 is directly connected, Ethernet0/0
      30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        30.0.0.0/8 is directly connected, Ethernet0/1
L        30.0.0.1/32 is directly connected, Ethernet0/1
D     40.0.0.0/8 [90/307200] via 30.0.0.2, 00:05:47, Ethernet0/1
R2(config)#ping 40.0.0.0
           ^
% Invalid input detected at '^' marker.

R2(config)#do ping 40.0.0.0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 40.0.0.0, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R2(config)#do ping 10.0.0.0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.0, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R2(config)#
```

R3

```
R3(config)#
R3(config)#do ping 20.0.0.0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 20.0.0.0, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R3(config)#do sh ip eigrp neighbor
EIGRP-IPv4 Neighbors for AS(10)
H   Address                 Interface              Hold Uptime   SRTT   RTO  Q
Seq
                                                   (sec)         (ms)       Cnt
Num
0   30.0.0.1                Et0/1                    10 00:07:45    5   100  0
7
R3(config)#do sh ip eigrp topology
EIGRP-IPv4 Topology Table for AS(10)/ID(40.0.0.1)
Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - reply Status, s - sia Status

P 10.0.0.0/8, 1 successors, FD is 332800
        via 30.0.0.1 (332800/307200), Ethernet0/1
P 30.0.0.0/8, 1 successors, FD is 281600
        via Connected, Ethernet0/1
P 40.0.0.0/8, 1 successors, FD is 281600
        via Connected, Ethernet0/0
P 20.0.0.0/8, 1 successors, FD is 307200
        via 30.0.0.1 (307200/281600), Ethernet0/1

R3(config)#ping 10.0.0.0
           ^
% Invalid input detected at '^' marker.

R3(config)#do ping 10.0.0.0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.0, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R3(config)#
```